

Memory-efficient tabular data using quadtree decomposition

Authors: Jared Carlson, Sean Couch, Brian W. O'Shea

Josh Briley - June 8, 2026

The Problem

Uniform Tabular Data

- Expensive functions are replaced by discretizing a domain and interpolating the tabulated function values.
- Uniform discretization is the most straightforward; but has issues.
 - Drastically varying interpolation errors -- under-refinement in some areas, over in others
 - Takes up a lot of computer memory
 - All cells perform the same interpolation method
- Either higher error than desired or larger memory cost than needed

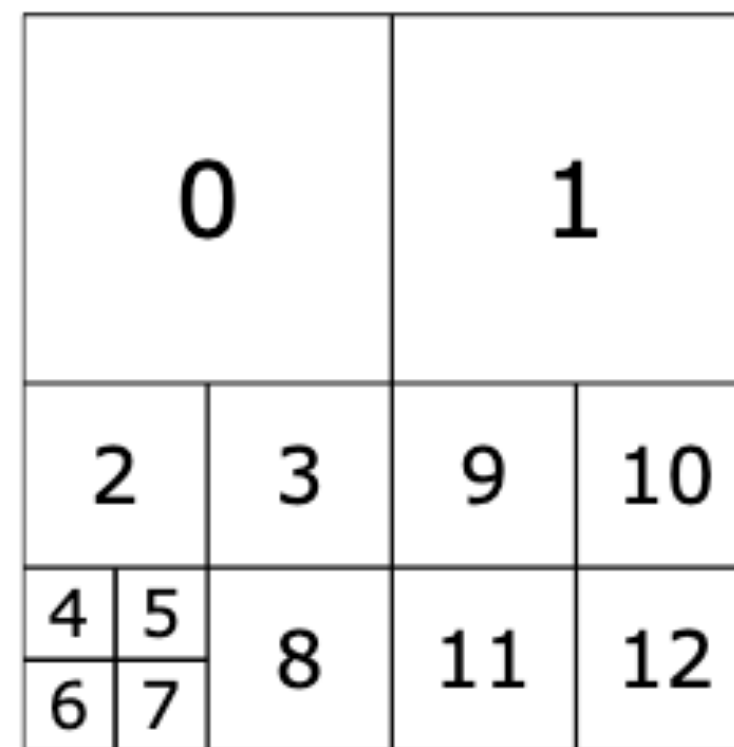
The solution? Quadtree
decomposition!

Birds-eye-view

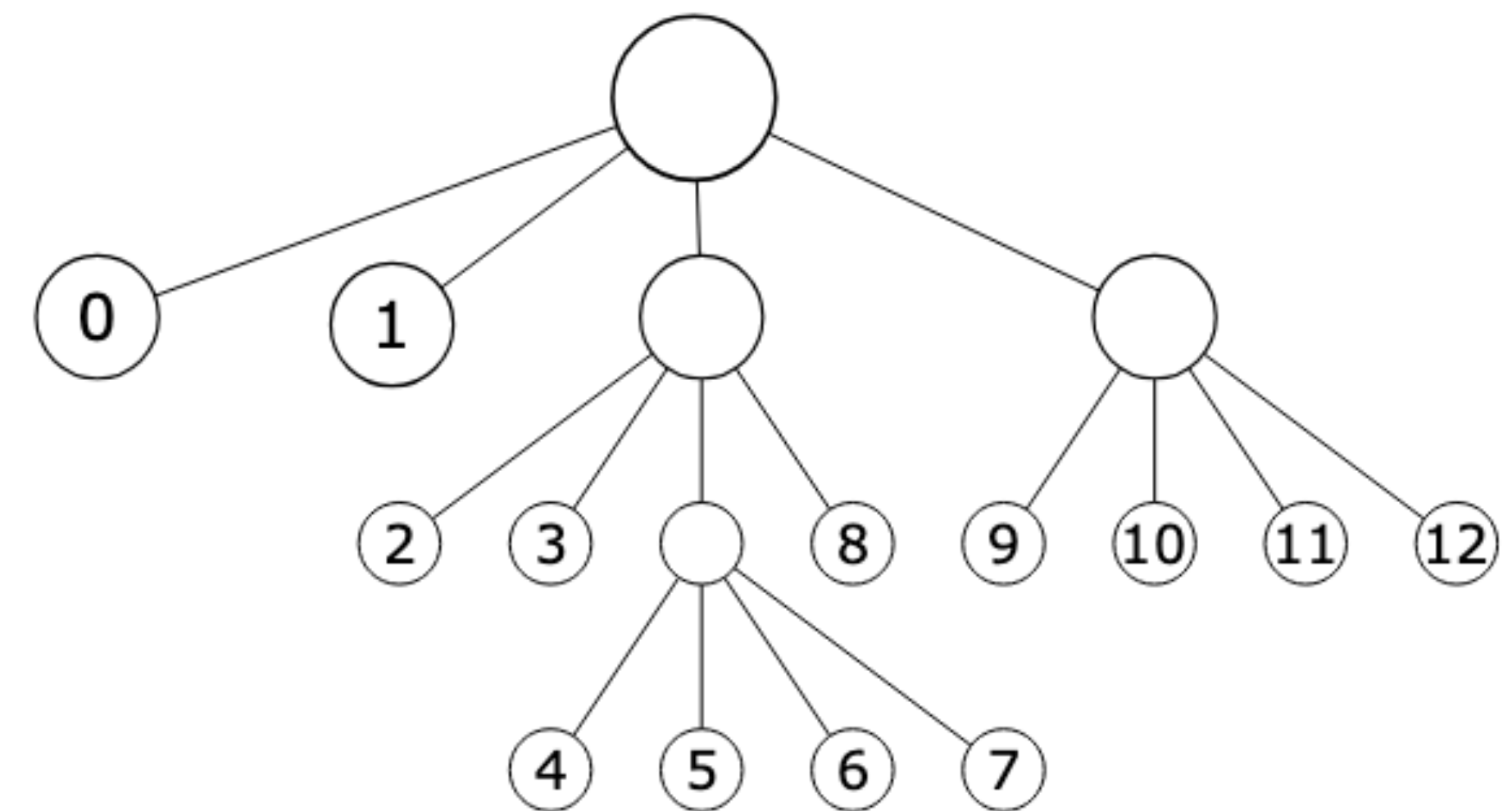
Quadtree decomposition

- The domain is initially split into four equal sized sub-domains (cells).
- The interpolation error is calculated in each cell.
- If that error is above a user-provided threshold (and a max-depth hasn't been reached), that cell is split again into fourths.
- This continues until the error is under the threshold or a max-depth has been reached.

2D Decomposition



Quadtree Representation



Calculating Error

Quadtree decomposition

- Training data is a uniform grid over the domain
 - Same resolution as TS00 as full-refinement
- Testing data is offset and a finer grid
 - Minimum 9 test points per cells
- Calculated L_1 , L_2 , L_∞ but only L_∞ was used in training.
 - Relative error

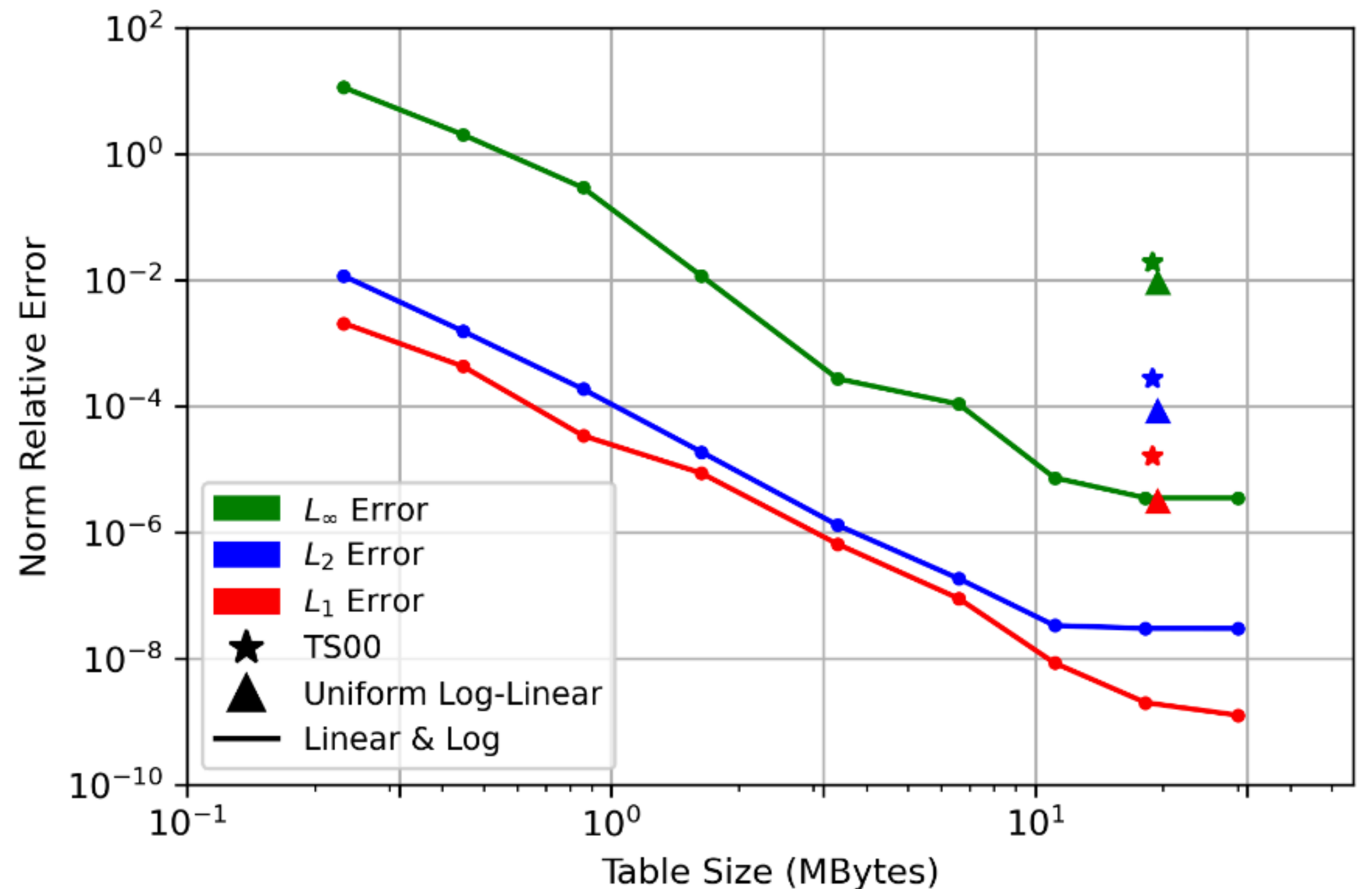
Proxy Problem

Helmholtz Free Energy

- Electron-Positron Helmholtz Free Energy Equation of State.
 - $f = f(\rho, T)$
 - Used in stellar plasma/hydrodynamics modeling
- Computationally expensive, evaluated multiple times per time-steps, changes orders of magnitude across domain
- TS00: Standard benchmark method. Uses uniform log-spaced grid.

Results

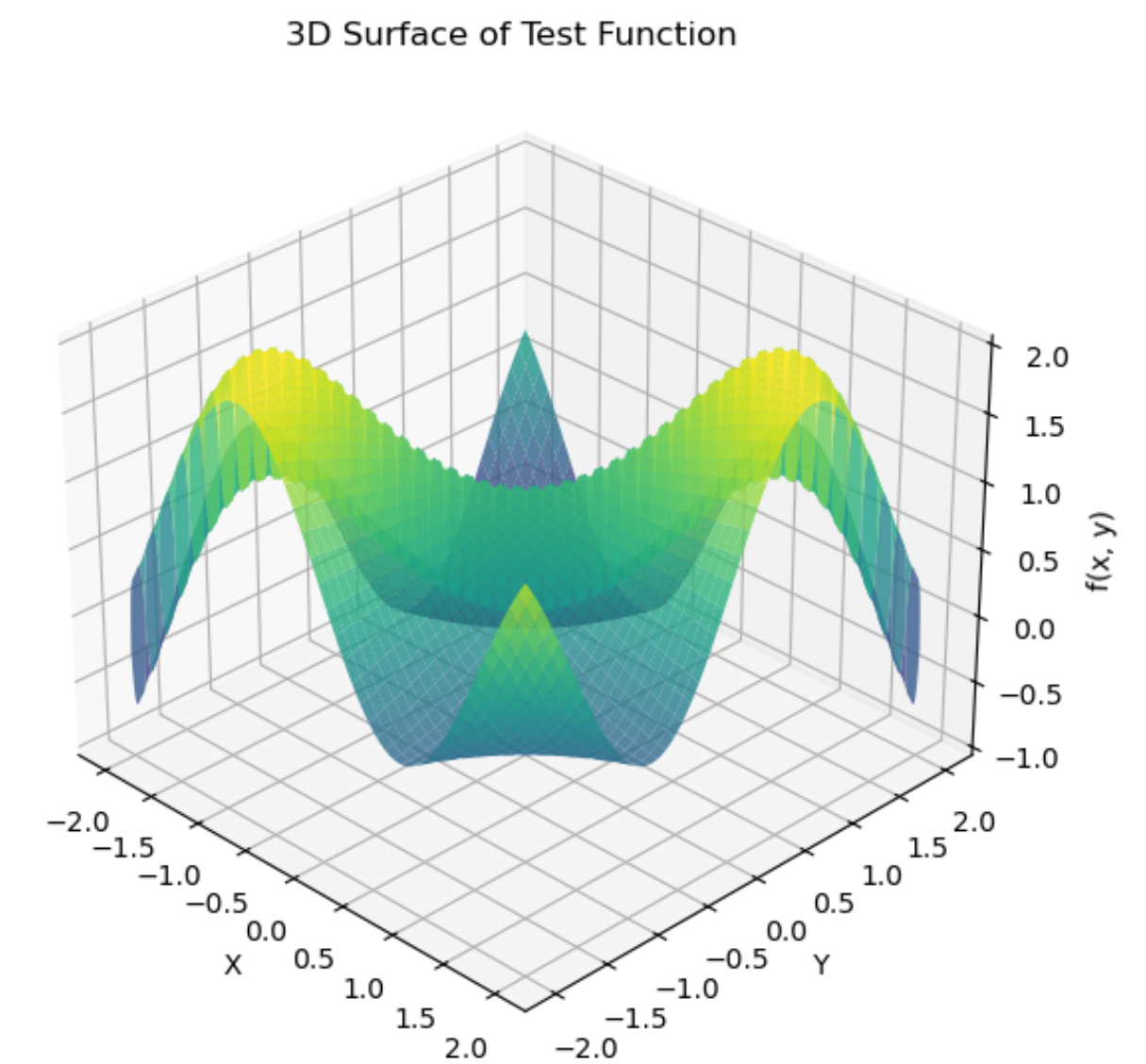
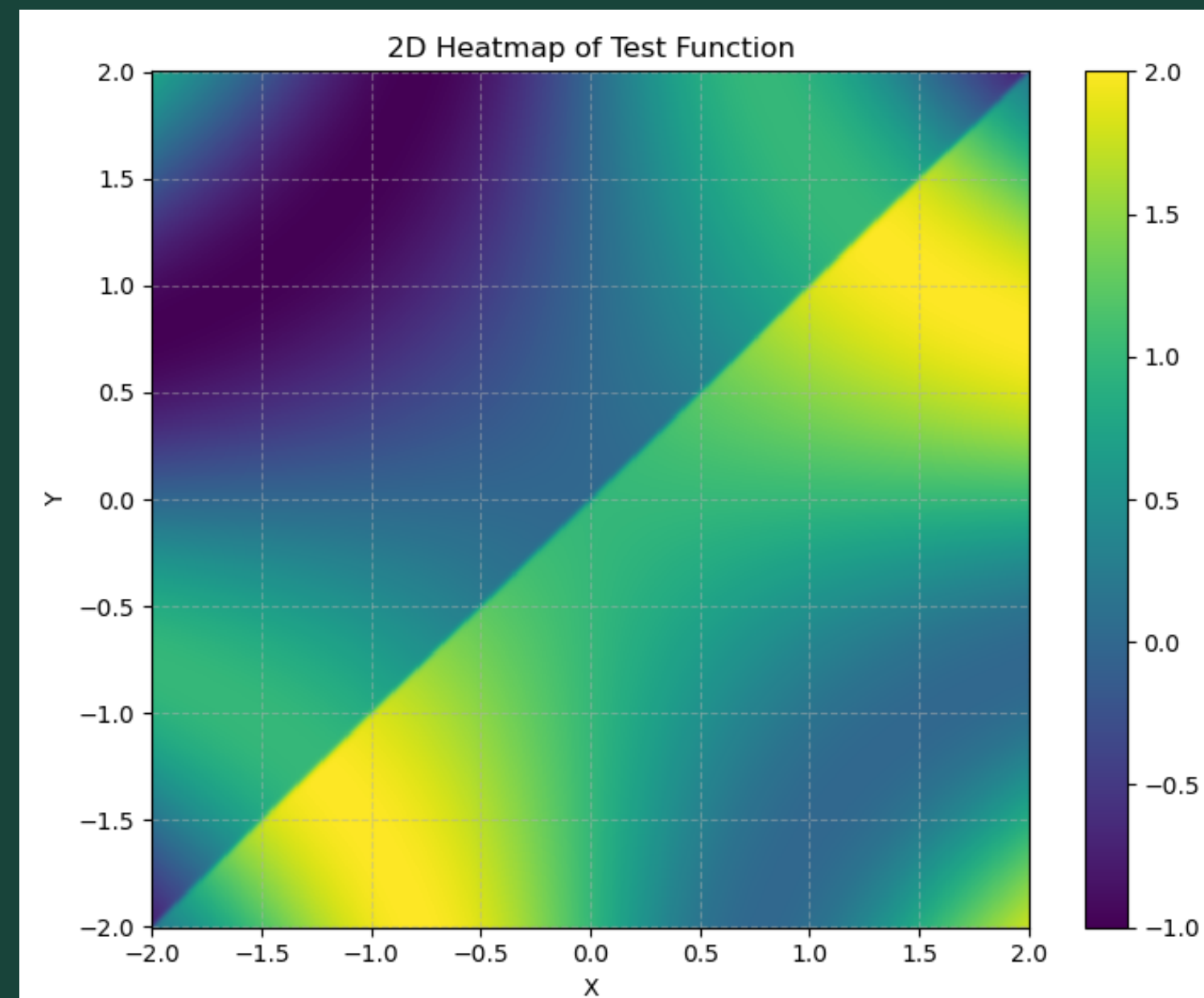
- Compared to the TS00
 - 20x savings in memory without loss of accuracy
 - Or, four orders of magnitude increase in accuracy for the same size table



My Implementation

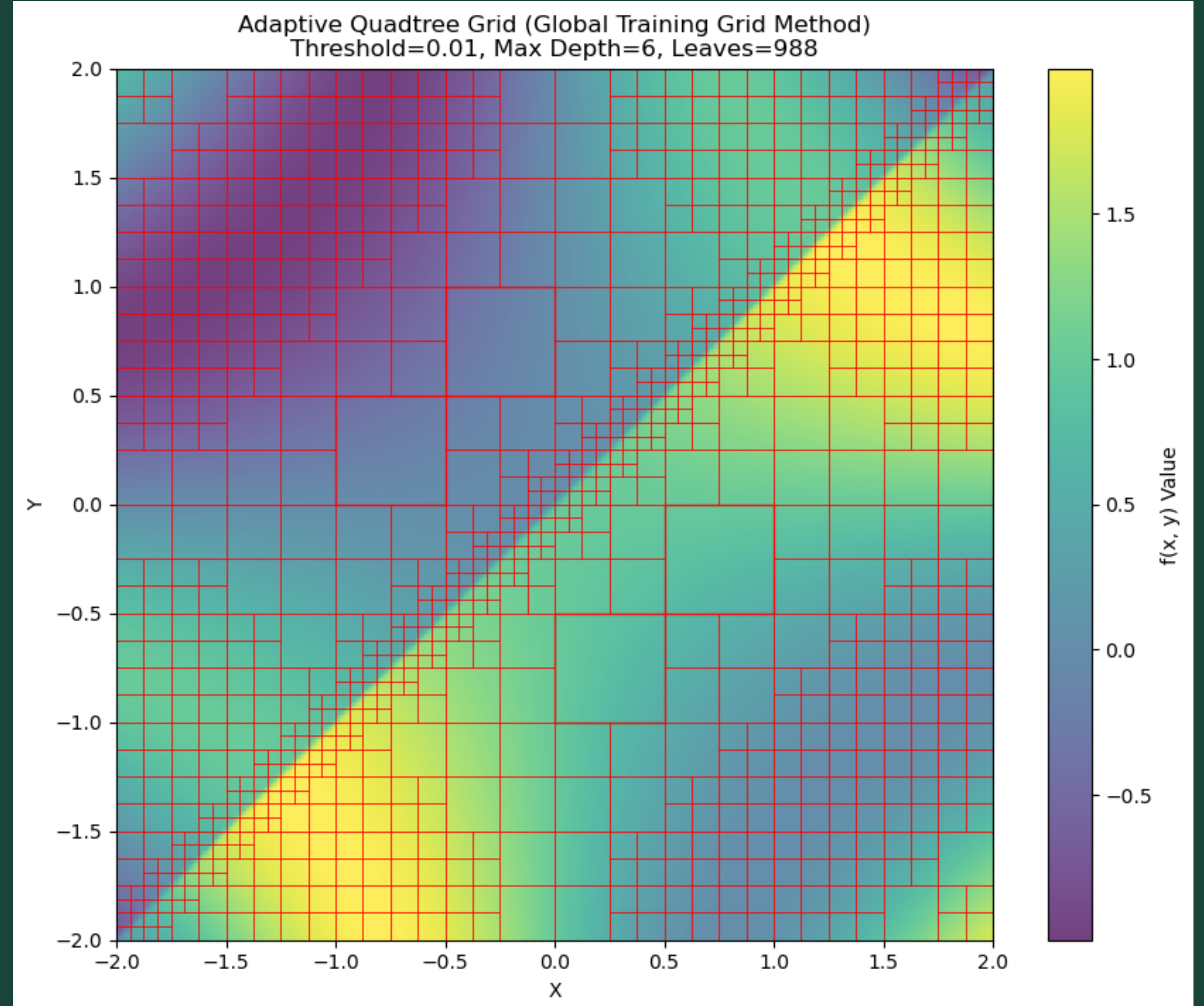
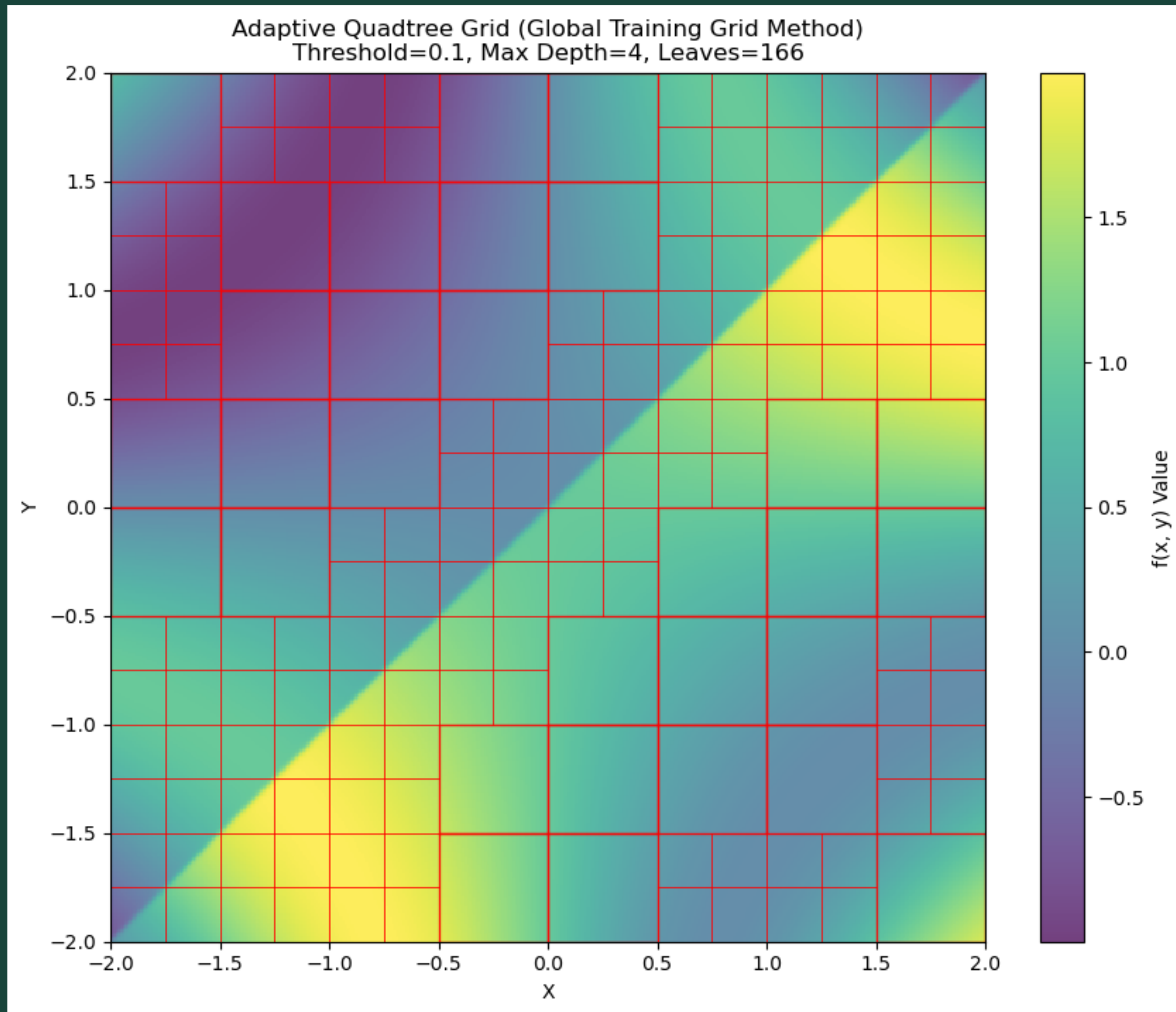
Proxy problem

- Proxy problem: $f(x, y) = \sin(xy) + \frac{1}{1 + e^{-100(x-y)}}$
- Linear interpolation
 - Uniform model throughout
- High-res image ->



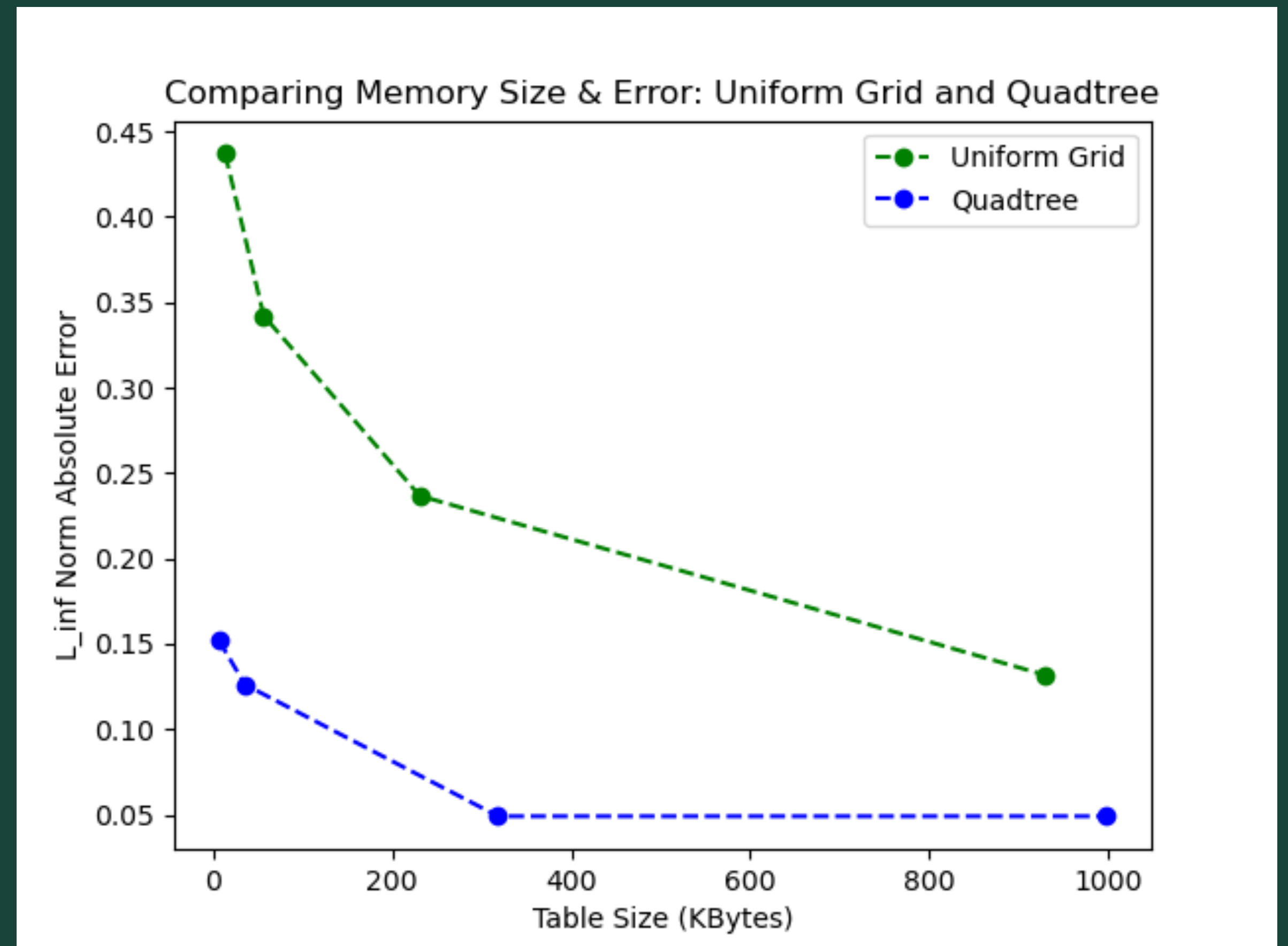
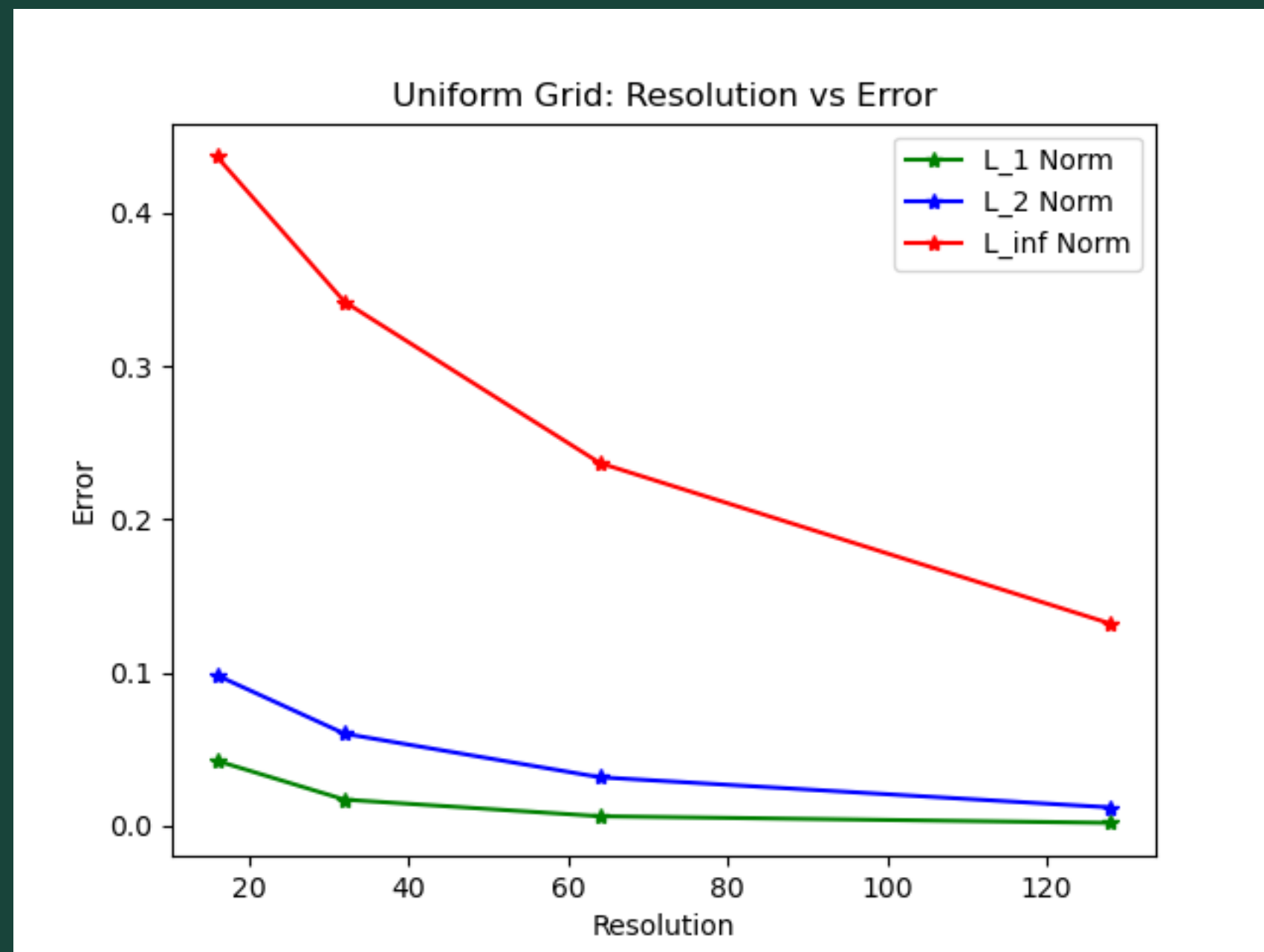
My Implementation

Quadtree Decomposition



My Implementation Results

- Quadtree: $\sim 6\text{KB} = \sim 0.15 L_\infty$
- Uniform: $\sim 929\text{KB} = \sim 0.13 L_\infty$



Things I am confused on

- Hybrid model selection
 - Linear-space model class
 - Log-space model class
- Memory compact mapping
 - Morton coding
- Specifics of testing data generation

Next-steps & relevance to me

- Increase to higher dimensions.
- Preserve discontinuities.
- Applies to multi-physics simulations.
- Section II.3.5 of the HighZ proposal.